
JHelioviewer: Coronagraph & PUNCH Preview

Installing and using the new features

Compiled by Chris Gilly, NWRA — a field guide for colleagues

What this is

This is a preview build of **JHelioviewer** (JHV), the open-source solar image browser from the Helioviewer Project. It carries a handful of new tools — a Sun-centered coronagraph view, a way to build “virtual coronagraphs” from ordinary disk images, support for NASA’s PUNCH mission data, and a radial enhancement filter. In essence, you are getting an early look.

Everything here is being contributed back to the official project and is working its way through review (the pull requests are listed on the last page). This build simply lets us use the features now, before the formal release lands. It is unofficial and lightly tested; please send problems to me, not to the JHV team.

1. Installing

JHV runs on macOS, Windows, and Linux. You need two things: a recent Java runtime, and the build itself.

Step 1 — Install Java (version 25 or newer)

JHV is written in Java, so it needs a Java runtime to start. The free Eclipse Temurin distribution works well.

- **Any system:** download “Temurin 25” (JRE or JDK) from adoptium.net and run the installer.
- **macOS with Homebrew:** open Terminal and run `brew install openjdk@25`.
- **Check it worked:** in a terminal, `java -version` should report 25 or higher.

Step 2 — Download the build

Download the single .zip I shared with you. If it is posted as a GitHub release, it lives at github.com/GillySpace27/JHelioviewer-SWHV/releases (the asset named like *JHelioviewer-PUNCH-preview.zip*). It is about 180 MB, because it carries the program and all its supporting libraries for every platform in one bundle.

Step 3 — Unzip and run

Unzip it somewhere convenient (your Desktop or Applications folder is fine), then open the folder and start it:

- **macOS:** double-click `run.command` (or run `./run.command` in a Terminal opened in this folder).
- **Linux:** run `./run.sh` in a terminal in this folder.
- **Windows:** double-click `run.bat`.

The launcher locates your Java 25 for you — including a Homebrew install that is not on your path. If it cannot find one, it prints how to install it.

If it will not open

“A newer Java is required.” Your default Java is older than 25. Re-check Step 1; you may have an older Java first on your path.

macOS: “cannot be opened, unidentified developer.” This is the normal Gatekeeper warning for downloaded files. Right-click `run.command` and choose **Open** (once), or in Terminal run `xattr -dr com.apple.quarantine` on the unzipped folder.

2. The new features

Each tool below has a short “what it does” and a “how to use it.” You do not need to read them in order; jump to whatever you came for.

PUNCH data

PUNCH is NASA’s four-spacecraft mission that images the faint solar wind across a very wide field. This build can pull PUNCH images straight from the public archive and play them as a movie.

How: open **File > New PUNCH Layer...** Choose a level, a product (the PTM and CTM “trefoil mosaics” are the high-cadence ones), a time range and a cadence, then **Search**, select the frames you want, and **Add**.

Two things are worth knowing. First, the **Version** box: PUNCH data is reprocessed over time (v0k, v0l, and so on), and mixing two processing versions in one movie makes the brightness jump frame to frame. “Latest” keeps the newest single version, or you can name one; either way a movie is never a mix. Second, the frames arrive already in the PUNCH color scheme, and a shared brightness range is computed in the background so the movie plays smoothly instead of flickering. A per-layer refresh button re-checks the archive for newer frames.

The coronagraph view (PowerDisk)

A normal coronagraph image crowds all of the faint outer corona into a thin rim at the edge. **PowerDisk** re-stretches the radial axis so that outer corona gets room to breathe. Basically it is a single knob that runs from an ordinary linear view, through a logarithmic one, to an inverse view that pulls infinity in to a finite edge.

How: click the **Projection** button in the top toolbar (the cube icon) and choose **PowerDisk**. A “p” slider appears in that same dropdown: $p = 1$ is linear, $p = 0$ is logarithmic (the default), and $p = -1$ is inverse. A second slider sets how much radius the view shows. Note that the disk itself is never warped — only the corona beyond the limb is re-stretched, so the Sun does not smear.

Building a virtual coronagraph

This is the showcase. You can stitch an on-disk image and a wider image of the same wavelength together so that a single instrument behaves like a coronagraph — the disk from one layer, the warped corona from the other.

- 1 Load two layers of the same wavelength — for example AIA 171 (the disk) and SUVI 171 (which sees a little farther out).
- 2 Switch the projection to **PowerDisk**.

- 3 On the disk layer, use the **Mask** slider in its layer options to show only the disk (mask away the corona).
- 4 On the wider layer, mask the disk away instead, and turn on its radial scaling with the small toggle in its **Filter** options — a **circle** icon means the radial scaling is on, a **rectangle** means it is held flat.
- 5 The two now read as one continuous disk-plus-stretched-corona image: a virtual coronagraph.

Why the toggle exists: in PowerDisk, ordinary disk images are held flat by default so their disk is not smeared. The circle/rectangle toggle lets you deliberately opt a disk image into the warp for exactly this trick. It is greyed out unless you are in the PowerDisk view, where it actually does something.

RHEF: bringing out the faint corona

The corona gets dramatically fainter with height, so its outer structure tends to vanish into black. The **Radial Histogram Equalizing Filter** (RHEF) flattens that falloff out, so faint outer features become visible without blowing out the bright inner corona. Sort of like auto-leveling the brightness separately at every radius.

How: in a layer's **Filter** options, set the filter to **RHEF**. If the result looks too harsh, the **Upsilon** control (the “Y” button) softens the shadows and the highlights independently.

Grid styling

The coordinate grid is now adjustable, which helps when it competes with your imagery. In the **Grid** layer's options you will find a color picker and sliders for line opacity, label opacity, line width, and label size; in the disk view there is also a slider that swings the radius labels to any angle, so you can move them off the part of the image you care about. Fading the grid to a light overlay while keeping the labels crisp is the common move.

3. Good to know

This is a working preview, not a finished release; the official JHV will eventually carry these features in a better-tested form. A couple of honest rough edges: only PUNCH version v0k is public at the moment, so the Version box has little to choose from until the next release; and the disk radial-range slider can still feel cramped at its low end if you also have a very wide-field layer loaded. Neither gets in the way of the features above.

Feedback is welcome and genuinely useful while these are in review. Send it to me, or comment directly on the pull requests below.

The pull requests (for the curious)

These are the changes as submitted to the Helioviewer Project, at github.com/Helioviewer-Project/JHelioviewer-SWHV:

#325 PowerDisk projection, radial mask, the flat-override toggle, the radial-range slider • #326 grid color / opacity / width and label size / angle • #327 the RHEF filter and Upsilon • #328 the New PUNCH Layer source and colormap • #329 the PUNCH refresh button • #330 the shared display range that keeps PUNCH movies from strobing.

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